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NPS/001 /012 – Specification for Cast Resin Pole Mounted Auxiliary Supply Transformers

1. Purpose

The purpose of this document is to provide a technical specification for the procurement of cast resin pole mounted auxiliary supply transformers for use on our 11, 20 & 33kV distribution system.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NEVT1	March 2002	Specification covering the supply of pole mounted auxiliary supply transformers.

2. Scope

This specification details the technical requirements relating to the supply of overhead line 11, 20 or 33kV cast resin voltage transformers (CRVT's) that are utilised primarily to provide 200VA, 230V AC auxiliary power supplies to pole mounted auto reclose circuit breakers.

3. Technical Requirements

3.1 Compliance with Standards

The equipment shall be designed to comply with the requirements of BS EN 60044-2 and with the Electromagnetic Compatibility (EMC) Regulations.

3.2 General Mounting Requirements

The CRVT's shall be securely mounted onto aluminium mounting plates and be suitable for attachment to wood poles.

3.3 Equipment ratings

3.3.1 Equipment ratings

The CRVT's are required for use on a three phase, 50Hz, distribution system with nominal system voltages of 11, 20 or 33kV distribution systems.

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The following table provides minimum requirements for the CRVT's

Nominal System Voltage (kV)	11kV	20kV	33kV
Highest voltage for equipment Um (r.m.s) (kV)	12	24	36
Secondary Output Voltage (V)	230	230	230
Primary Winding - Rated Power Frequency withstand voltage (kV)	28	50	70
Secondary Winding - Rated Power Frequency short duration withstand voltage (kV)	3kV for 1 Min	3kV for 1 Min	3kV for 1 Min
Lightning impulse withstand voltage (BIL) (kV)	125	170	200
Power rating (VA)	200	200	200
Voltage Factor	1.2 Um continuously	1.2 Um continuously	1.2 Um continuously
*Min Nominal specific creepage distance (BS EN 60044-2 Table 8) (mm)	275	500	825
**Min separation between primary winding terminals & Primary terminals to earth (mm)	210	280	280

* Based on 25mm/kV form IEC 815 class III

** Based on BS EN 60076-2 Table 5

3.3.2 System Earthing

The equipment shall be suitable for use on a distribution system with the neutral Solidly Earthed, Impedance Earthed or via a Resonant Earthing system.

3.4 Normal Service Conditions

The CRVT's shall be capable of operation within the following service conditions:

- Ambient temperature range of -25°C to +40°C where the average value of the ambient air temperature, measured over a period of 24h, does not exceed 35°C.
- Solar radiation up to a level of 1000 W/m² (on a clear day at noon) should be considered.
- The ambient air may be polluted by dust, smoke, corrosive gases, vapours or salt. However the pollution levels will not be required to exceed the pollution levels given in BS EN 60044-2, table 8.
- The wind pressure will not be required to exceed 700 Pa (corresponding to 35 m/s wind speed).
- Account should be taken of the any detrimental effects created by condensation and precipitation.
- The maximum installation altitude will not be required to exceed 1000m.

3.5 LV Terminal Enclosure

The design of the CRVT shall incorporate a waterproof LV terminal enclosure with a removable lid. All fixings used to secure the lid to the enclosure shall be captivated.

LV terminals marked X1 and X2, located within the enclosure shall be suitable for the termination of 2.5mm PVC insulated conductor. The unit shall incorporate a suitable removable link between an internal earth terminal and the X1 terminal on the LV winding (- polarity side).

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The earth terminal located within the enclosure shall be effectively bonded to the transformer mounting plate.

The enclosure shall incorporate a cable entry point suitable for the termination of an M20 cable gland.

3.6 Rating and Connection Plates

Transformers shall be fitted with a rating plate generally in accordance with Fig 1 ENA TS 35-1 with the addition of the following:-

- a) Rated output and accuracy class
- b) Rated Voltage factor and corresponding time
- c) Class of insulation
- d) Winding ratio

A connection plate is required to show the winding connections and shall be generally in accordance with Fig 2 to Fig 7 of ENA TS 35-1. The connection plate may be part of the rating plate.

Rating and connection plates shall be of durable and non-corrodible material and shall be securely fixed to the mounting plate

3.7 Terminal Markings

Terminals shall be clearly marked and identified in accordance with BS EN 60044-2 Section 11.2.

3.8 Losses

CRVT's shall provide the following typical losses:

Iron Loss	36 (W)
Copper	20 (W)

These loss values detailed above shall be taken as the guaranteed losses and the tolerances shall be measured in accordance with IEC 60076.

3.9 Testing

3.9.1 Routine Tests

Routine tests shall be carried out on all transformers as specified in BS EN 60044-2 clause 9.

Partial discharge test procedure shall be carried out as described in clause 9.2.4 & 9.2.4.3 using procedure 'B' with maximum allowable discharge levels compliant with Table 7 of BS EN 60044-2.– See Note.

Where it is deemed appropriate and agreed by CE Electric, the test procedure detailed in the Canadian standard CAN3-C13-M83 with its associated 50pc PD limit may be utilised as an alternative to the IEC 60044-2 PD test requirements.

Note:

Note 3 of BSEN 60044-2 Table 7 states, "When the rated voltage transformer is considerably lower than its highest system voltage U_m , lower pre-stress voltages and measuring voltages may be agreed between the manufacturer and the purchaser".

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Where it is deemed appropriate and agreed by CE Electric the following reduced value test procedure may be applied on 20kV units.

Apply a pre-stress voltage of 80% of U_m (40kV for 60 secs) followed by the partial discharge measurement at 25.2kV ($1.2 \times 21kV$) instead of 28kV.

3.9.2 Type tests

Type tests in addition to routine tests shall be carried out on the first unit of any design. The tests shall be carried out in accordance with BSEN 60044-2 clause 8.

Sound pressure level tests shall be carried out in accordance with BS EN 60076-10 as part of the type test program. The sound power level derived from measurements made in accordance with BS EN 60076-10 shall not exceed 45 (dBA).

3.10 Flux Density

The maximum flux density in any magnetic part shall not exceed 1.9 Tesla with 1.1 U_m applied at a frequency of 47 Hz. Transformers shall not over flux under these conditions.

3.11 Over Voltage Protection

No over voltage protection other than the increased BIL requirements for these transformers has been specified.

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4. References

4.1 External Documentation

Reference	Title
BS EN 60044-2:1999	Instrument transformers. Inductive Voltage Transformers
BS EN 60076 – 10:2001	Power Transformers
ENA TS 35-1: 1995	Distribution Transformers (from 16kVA to 1000kVA)

4.2 Internal documentation

Reference	Title
N/A	N/A

5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	12/01/2006

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G.Bolden	Purchasing Manager	G.Bolden	12/01/2006

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Standards Manager	G Hammel	11/01/2006

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	20/01/2006

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards & Assessment Manager	C.Holdsworth	12/01/2006

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	13/01/2006

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Schedule A

SELF CERTIFICATION CONFORMANCE DECLARATION

Cast Resin Pole Mounted Auxiliary Transformers covered by BS EN 60044 – 2 shall comply with this technical specification and the latest issues of the relevant international and British Standards.

BE EN 60044 - 2 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in BS EN 60044 - 2 and the clauses of the aforementioned Standards relevant to Cast Resin Pole Mounted Auxiliary Transformers for use on the distribution networks of NEDL and YEDL.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Manufacturer:

Product Reference:

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENATS' as appropriate

Name:

Signature:

Date:

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Schedule A Table

Cast Resin Pole Mounted Auxiliary Transformers

• BS EN 60044 - 2			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
3	General Requirements		
4	Normal and Special Service Conditions		
5	Ratings		
6	Design requirements		
8	Type Tests		
9	Routine Tests		
10	Special Tests		
11	Markings		
• NPS/001/012			
3.3.1	Minimum Requirements		

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Schedule B – Logistical Requirements

1.0 Protection and Packaging

The equipment shall be suitably packaged and protected with each unit marked with the following information:

- (i) Manufacturers name.
- (ii) Manufacturers product type and unique unit serial number.
- (iii) Description of item.
- (iv) Date of supply.
- (v) Company order number.
- (vi) Purchasing Companies Stock Catalogue Number.
- (vii) Weight of each unit.

2.0 Test Certificates

All cast resin pole mounted auxiliary transformers shall be supplied complete with a copy of the routine tests carried out on the equipment during its manufacture. The test cards shall be enclosed in a waterproof document holder and be attached to the unit.